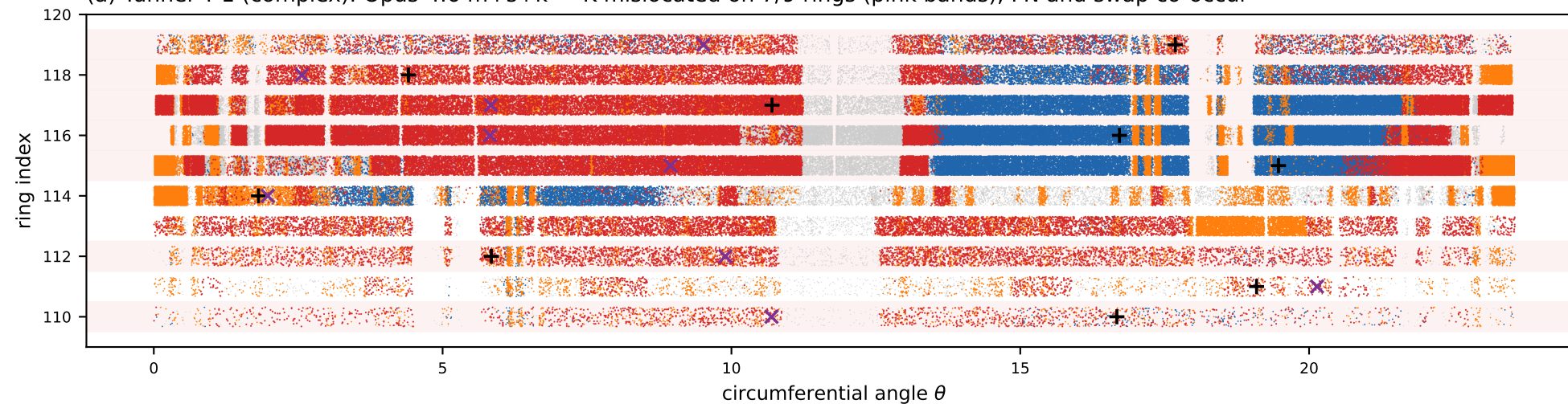
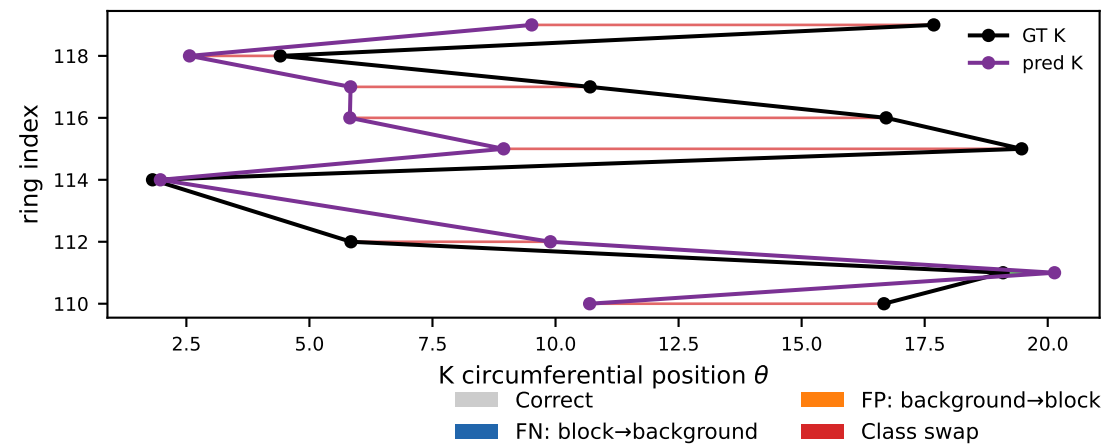


Complex tunnel: jumping K-anchor drives failure; class swaps add on top

(a) Tunnel 4-1 (complex): Opus-4.6 m+s+k — K mislocated on 7/9 rings (pink bands); FN and swap co-occur



(b) Predicted K jumps ring-to-ring while GT K follows staggered assembly



(c) Accuracy collapses when K offset > 0.5 sector (mislocated mean acc 0.09)

